

Publication 1

Title	Enhancing customer retention in telecom industry with machine learning driven churn prediction. Scientific Reports, 14, 13097.
Harvard reference	Sikri, A., Jameel, R., Idrees, S.M. and Kaur, H. (2024). Enhancing customer retention in telecom industry with machine learning driven churn prediction. Scientific Reports, 14, 13097.
Link	https://doi.org/10.1038/s41598-024-63750-0
Modelling	Perceptron, MLP, Naive Bayes, Logistic Regression, KNN, Decision Tree, Gradient Boosting and XGBoost. The paper compares traditional resampling with a ratio-based balancing technique.
Preprocessing techniques	Data balancing was a central preprocessing step. The study compares the proposed ratio-based approach with common over-/under-sampling, followed by supervised classification.
Datasets used	A publicly available customer-churn dataset downloaded from Kaggle; the study focuses on churn classification with imbalanced classes.
Key takeaways	1) Class imbalance can materially affect churn metrics. 2) Balancing strategy should be documented and tested. 3) Ensemble methods such as gradient boosting/XGBoost are useful candidates for nonlinear churn patterns.

Publication 2

Title	Customer churn prediction in telecom using machine learning in big data platform. Journal of Big Data, 6, 28.
Harvard reference	Ahmad, A.K., Jafar, A. and Aljoumaa, K. (2019). Customer churn prediction in telecom using machine learning in big data platform. Journal of Big Data, 6, 28.
Link	https://doi.org/10.1186/s40537-019-0191-6
Modelling	Decision Tree, Random Forest, Gradient Boosted Machine and XGBoost. The study also used feature engineering/selection and evaluated models with AUC.
Preprocessing techniques	Removed identifiers and highly incomplete/constant variables, handled missing values, created aggregated behavioural features, and tested several class-rebalancing scenarios.
Datasets used	Large SyriaTel customer dataset covering multiple operational sources and nine months of customer information before a prediction baseline.
Key takeaways	1) Feature engineering can be more valuable than simply increasing model complexity. 2) Identifiers should not be used as predictive features. 3) AUC is useful when the positive churn class is much smaller than the active class.

Publication 3

Title	Machine Learning–Based Customer Churn Prediction in Telecommunication Industry. Applied Computational Intelligence and Soft Computing, 2026, 8496186.
Harvard reference	Mettle, H.N-A., Ayerko, E.H. and Appiahene, P. (2026). Machine Learning–Based Customer Churn Prediction in Telecommunication Industry. Applied Computational Intelligence and Soft Computing, 2026, 8496186.
Link	https://doi.org/10.1155/acis/8496186
Modelling	Decision Tree, Random Forest, XGBoost and Logistic Regression. The study also reports a tuned soft-voting ensemble and evaluates accuracy, precision, recall, F1-score and ROC-AUC.
Preprocessing techniques	Converted/cleaned TotalCharges, handled missing/inconsistent records, encoded categorical variables, and used a tuned modelling workflow. The study retained 5,634 records after preprocessing.
Datasets used	IBM Telco Customer Churn dataset; original dataset contains 7,043 records and 21 columns.
Key takeaways	1) Evaluate more than accuracy because churn is imbalanced. 2) Compare an interpretable linear baseline with an ensemble tree model. 3) Use ROC-AUC, precision, recall and F1-score together when assessing churn risk.

Public Dataset Description

Name	IBM Telco Customer Churn
Harvard reference	IBM (2019) Telco Customer Churn sample data.
Link	https://www.ibm.com/docs/en/cognos-analytics/12.1.x?topic=samples-telco-customer-churn
Number of instances	7,043 customer records
Number of feature columns	21 columns in the raw file, including customerID and the Churn target; 20 predictor columns before removing the identifier.
Number of classes	2 (Yes/No)
Target range	Not applicable; this is a binary classification problem.
Origin	IBM sample data for a fictional telecommunications company. The Churn field indicates whether the customer departed within the last month.